

SHANTANU GHOSH

Ph.D. Candidate, Electrical and Computer Engineering, Boston University

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RESEARCH INTERESTS

Trustworthy machine learning; robustness and generalization; multimodal learning; explainable artificial intelligence; vision-language models for medical imaging; radiology report generation; breast cancer risk prediction and chest radiograph analysis.

EDUCATION

Boston University Jan 2023–Dec 2026 expected
Ph.D., Electrical Engineering. Thesis: *Interpretable Medical AI with Vision-Language Alignment*. Advisor: [Kayhan Batmanghelich](#). Committee: [Brian Kulis](#), [Wenchao Li](#), [Clare Poynton](#).

University of Pittsburgh Aug 2021–Dec 2022
Ph.D. student, Intelligent Systems Program. Transferred to Boston University following advisor move.

University of Florida Aug 2019–May 2021
M.S., Computer Science. GPA: 3.88/4.00. Advisor: [Mattia Prosperi](#).

Institute of Engineering and Management Aug 2008–May 2012
B.Tech., Computer Science. Kolkata, India.

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Graduate Research Assistant, Boston University Jan 2023–present
Advisor: [Kayhan Batmanghelich](#); clinical collaborator: [Clare Poynton](#).

- Developed LADDER, a language-driven slice discovery and error rectification method for vision classifiers; accepted at ACL 2025 Findings.
- Developed Mammo-CLIP, a vision-language foundation model for 2D mammography; accepted at MICCAI 2024 as an early accept paper.
- Applied mixtures of interpretable models for efficient transfer learning in chest radiograph classification; accepted at MICCAI 2023 as an early accept paper.

Graduate Student Researcher, University of Pittsburgh Aug 2021–Dec 2022
Advisor: [Kayhan Batmanghelich](#); collaborator: [Forough Arabshahi](#).

- Developed an iterative algorithm to extract mixtures of interpretable models from black-box classifiers using human-understandable concepts and first-order-logic explanations; accepted at ICML 2023.

Graduate Research Assistant, University of Florida Jan 2020–Jul 2021
Advisors: [Kevin Butler](#), [Jiang Bian](#), and [Mattia Prosperi](#).

- Developed DPN-SA, PSSAM-GAN, and DR-VIDAL for propensity score estimation, counterfactual prediction, and treatment-effect estimation.
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INDUSTRY EXPERIENCE

United Imaging Intelligence (UII-America) May 2026–Aug 2026

Research Scientist Intern, UII Computer Vision. Mentor: [Ziyan Wu](#). Developing video LLMs for surgical video understanding.

Amazon Web Services, Optimus Team

May 2025–Aug 2025

Applied Scientist Intern. Mentor: [Ankan Bansal](#). Developed methods to detect and mitigate biases in multimodal LLM coding agents.

Amazon Web Services, Security Analytics and AI Research

Jun 2024–Sep 2024

Applied Scientist Intern. Mentor: [Mikhail Kuznetsov](#). Developed robust representation learning methods for systematic-error mitigation in tabular self-supervised models; publication at TRL@NeurIPS 2024.

Lexmark International India Pvt. Ltd.

Oct 2016–Jul 2019

Software Engineering Professional II. Developed the ISP component of [Publishing Platform for Retail](#).

Cognizant Technology Solutions India Pvt. Ltd.

Mar 2013–Sep 2016

Associate, Projects. Developed WCF web services using a contract-first service-oriented architecture.

PUBLICATIONS

Preprints

- **Mammo-FM: Breast-specific foundational model for Integrated Mammographic Diagnosis, Prognosis, and Reporting.** Shantanu Ghosh, Vedant Parthesh Joshi, Rayan Syed, Aya Kassem, Abhishek Varshney, Payel Basak, Judy Wawira Gichoya, Hari M. Trivedi, Imon Banerjee, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich. *arXiv*. 2026. [\[Paper\]](#)

Conference Proceedings

- **MedSyn2: Flexible Control of 3D CT Generation via Text and Semantically-Defined Segmentation Prompts.** Weicheng Dai, Chenyu Wang, Binxu Li, Shantanu Ghosh, Afroz Zandifar, Christina LeBedis, Kayhan Batmanghelich. *European Conference on Computer Vision (ECCV)*. 2026. [\[Paper\]](#)
- **PhyDiCT: Plug-and-Play CT Reconstruction from Sparse X-Rays via Differentiable Rendering and Strong Priors.** Weicheng Dai, Shantanu Ghosh, Kayhan Batmanghelich. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2026.
- **LADDER: Language-Driven Slice Discovery and Error Rectification in Vision Classifiers.** Shantanu Ghosh, Rayan Syed, Chenyu Wang, Vaibhav Choudhary, Binxu Li, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich. *Findings of the Association for Computational Linguistics (ACL)*. 2025. [\[Project\]](#) [\[Paper\]](#) [\[Code\]](#)
- **Semantic Consistency-Based Uncertainty Quantification for Factuality in Radiology Report Generation.** Chenyu Wang, Weichao Zhou, Shantanu Ghosh, Kayhan Batmanghelich, Wenchao Li. *Findings of the Association for Computational Linguistics: NAACL*. 2025. [\[Paper\]](#) [\[Code\]](#)
- **Mammo-CLIP: A Vision Language Foundation Model to Enhance Data Efficiency and Robustness in Mammography.** Shantanu Ghosh, Clare B. Poynton, Shyam Visweswaran, Kayhan Batmanghelich. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2024. Early accept, top 11%. [\[Project\]](#) [\[Paper\]](#) [\[Code\]](#)
- **Distilling BlackBox to Interpretable models for Efficient Transfer Learning.** Shantanu Ghosh, Ke Yu, Kayhan Batmanghelich. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2023. Early accept, top 14%. [\[Project\]](#) [\[Paper\]](#) [\[Code\]](#)
- **Dividing and Conquering a BlackBox to a Mixture of Interpretable Models: Route, Interpret, Repeat.** Shantanu Ghosh, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich. *International Conference on Machine Learning (ICML)*. 2023. [\[Project\]](#) [\[Paper\]](#) [\[Code\]](#)
- **DR-VIDAL: Doubly Robust Variational Information-theoretic Deep Adversarial Learning for Counterfactual Prediction and Treatment Effect Estimation.** Shantanu Ghosh, Zheng Feng, Jiang Bian, Kevin Butler, Mattia Prosperi. *American Medical Informatics Association (AMIA) Symposium*. 2022. Oral presentation. [\[Paper\]](#) [\[Code\]](#)

- **Anatomy-Guided Weakly-Supervised Abnormality Localization in Chest X-rays.** Ke Yu, **Shantanu Ghosh**, Zhexiong Liu, Christopher Deible, Kayhan Batmanghelich. *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*. 2022. [[Paper](#)] [[Code](#)]
- **Causal AI with Real World Data: Do Statins Protect From Alzheimer’s Disease Onset?.** Mattia Prosperi, **Shantanu Ghosh**, Zhaoyi Chen, Marco Salemi, Tianchen Lyu, Jiang Bian. *International Conference on Medical and Health Informatics (ICMHI)*. 2021.

Journal Articles

- **Anatomy-specific Progression Classification in Chest Radiographs via Weakly-Supervised Learning.** Ke Yu, **Shantanu Ghosh**, Zhexiong Liu, Clare Poynton, Christopher Deible, Kayhan Batmanghelich. *Radiology: Artificial Intelligence*. 2024. [[Paper](#)] [[Code](#)]
- **Propensity Score Synthetic Augmentation Matching using Generative Adversarial Networks.** **Shantanu Ghosh**, Christina Boucher, Jiang Bian, Mattia Prosperi. *Computer Methods and Programs in Biomedicine Update*. 2021. [[Paper](#)] [[Code](#)]
- **Deep Propensity Network using a Sparse Autoencoder for Estimation of Treatment Effects.** **Shantanu Ghosh**, Jiang Bian, Yi Guo, Mattia Prosperi. *Journal of the American Medical Informatics Association*. 2021. [[Paper](#)] [[Code](#)]

Peer-Reviewed Workshops

- **Distributionally robust self-supervised learning for tabular data.** **Shantanu Ghosh**, Tiankang Xie, Mikhail Kuznetsov. *Table Representation Learning Workshop, NeurIPS*. 2024. [[Paper](#)] [[Code](#)]
- **Tackling Shortcut Learning in Deep Neural Networks: An Iterative Approach with Interpretable Models.** **Shantanu Ghosh**, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich. *Workshop on Spurious Correlations, Invariance and Stability, ICML*. 2023. [[Paper](#)]
- **Bridging the Gap: From Post Hoc Explanations to Inherently Interpretable Models for Medical Imaging.** **Shantanu Ghosh**, Ke Yu, Forough Arabshahi, Kayhan Batmanghelich. *Workshop on Interpretable Machine Learning in Healthcare, ICML*. 2023. [[Paper](#)]

TEACHING

- Medical Imaging With AI (EC 500), Boston University • Guest Lecturer • Fall 2025
- Deep Learning (EC 523), Boston University • Teaching Assistant • Fall 2024
- Introduction to Software Engineering (EC 327), Boston University • Teaching Assistant • Fall 2023

STUDENT MENTORING

- [Rayan Syed](#), undergraduate student, Boston University • BU Outstanding Undergraduate Research Award.
- [Ho Cheung Aiden Wong](#), undergraduate student, Boston University • Next: M.S. Computer Science, Columbia University.
- [Alex Tang](#), undergraduate student, Dartmouth University.
- [Ryan Gupta](#), undergraduate student, Boston University.
- [Abhishek Varshney](#), master’s student, Boston University • Next: Machine Learning Engineer, Poshmark.
- [Akshat Gurbuxani](#), master’s student, Boston University • Next: Software Engineer, Semiotic Labs.

ACADEMIC SERVICE

Journal Reviewing

Transactions on Machine Learning Research; IEEE Transactions on Medical Imaging; Journal of Biomedical Informatics; Medical Image Analysis; Journal of the American Medical Informatics Association; Computer Methods and Programs in Biomedicine; Biometrical Journal; Information Fusion.

Conference and Workshop Reviewing

BMVC 2026; ECCV 2026; WACV 2025; ICCV 2025; ICML 2025; ICLR 2024–2026; AAAI 2024–2026; AISTATS 2025; NeurIPS 2023–2026; MICCAI 2024–2026; CVPR 2024–2026; CLeaR 2024–2025; GenAI4Health@NeurIPS 2024–2026; SCIS@ICML 2023; IMLH@ICML 2023.

HONORS AND AWARDS

- Full Travel Award, Symposium on Artificial Intelligence for Learning Health Systems (SAIL), 2026.
- Outstanding Reviewer, NeurIPS, 2024.
- Achievement Award, University of Florida graduate admission, 2019.
- Star Employee Award, Cognizant Technology Solutions, Q4 2013 and Q4 2015.

TALKS AND ABSTRACT PRESENTATIONS

- *Text-Grounded Interpretability for Mammography Risk Prediction via Vision-Language Alignment*. SAIL 2026; International Breast Density and Cancer Risk Assessment Workshop, 2025.
- *A Domain-Specific Foundation Model Surpassing Generalist AI for Integrated Mammographic Diagnosis, Prognosis and Reporting*. Annual Boston Medical Imaging Workshop, MIT CSAIL, 2025.
- *Divide and Conquer: Carving Out Concept-based Models out of BlackBox for More Efficient Transfer Learning*. MedAI Group, Stanford University, 2024; ISP AI Forum, University of Pittsburgh, 2023.
- *DR-VIDAL for Counterfactual Prediction and Treatment Effect Estimation*. Oral talk, AMIA Annual Symposium, 2022.

TECHNICAL SKILLS

Python; PyTorch; TensorFlow; scikit-learn; C/C++; Java; C#/.NET; JavaScript; HTML/CSS; Angular; Node.js; Oracle; MS SQL Server.